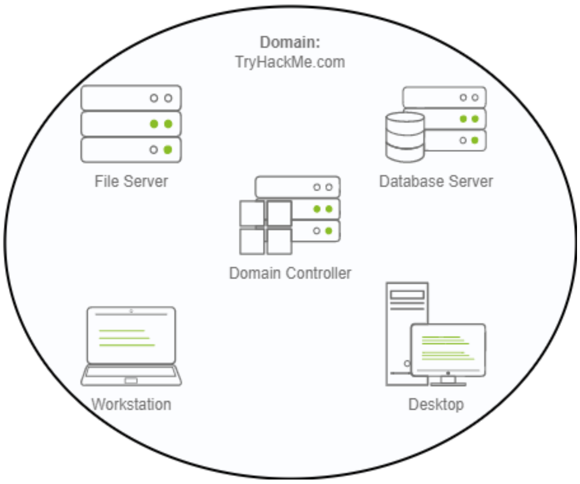
Active Directory Basics

Simplifies the management of devices and users within a corporate environment

Windows Domains

It's a group of users and computers under administration of a given business. -Main goal behind domain is to centralize the administration of a common components of a window computer network in a single repository called **Active Directory (AD).**

Server that runs the active directory services is known **Domain Controller (DC)**



Advantages:

- **Centralised identity management:** All users across the network can be configured from Active Directory with minimum effort.
- **Managing security policies:** You can configure security policies directly from Active Directory and apply them to users and computers across the network as needed

Active Directory Domain Service (AD DS)

It's a catalogue that holds the information of all the objects that exist on the network.

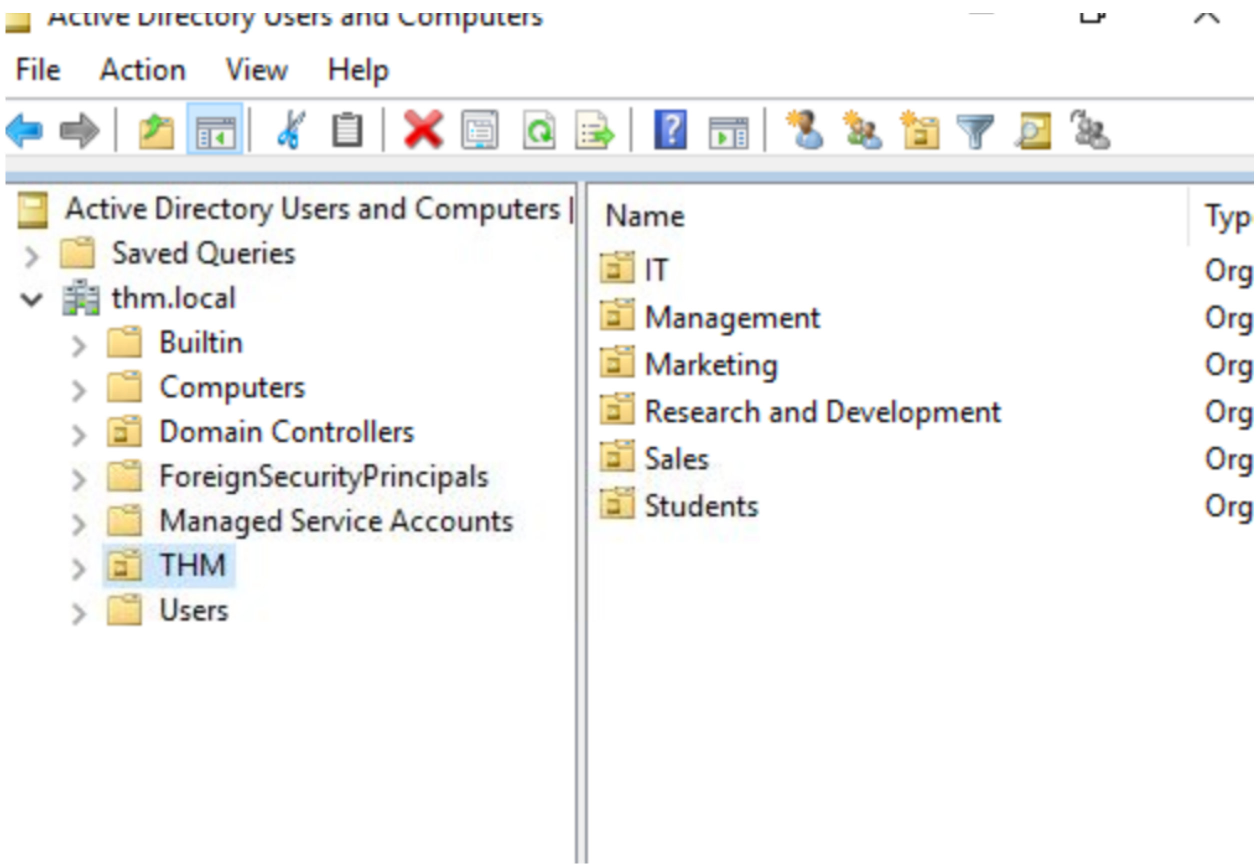
Objects are:users,groups,machines,printers.

-Users also known as **(Security principals)**-they can authenticated by the domain and can be assigned privileges over resources eg files or printers.

User can be people, services

-Machine for every computer that join active directory domain machine should be created.

Security Group	**Description**
Domain Admins	Users of this group have administrative privileges over the entire domain. By default, they can administer any computer on the domain, including the DCs.
Server Operators	Users in this group can administer Domain Controllers. They cannot change any administrative group memberships.
Backup Operators	Users in this group are allowed to access any file, ignoring their permissions. They are used to perform backups of data on computers.
Account Operators	Users in this group can create or modify other accounts in the domain.
Domain Users	Includes all existing user accounts in the domain.
Domain Computers	Includes all existing computers in the domain.
Domain Controllers	Includes all existing DCs on the domain.



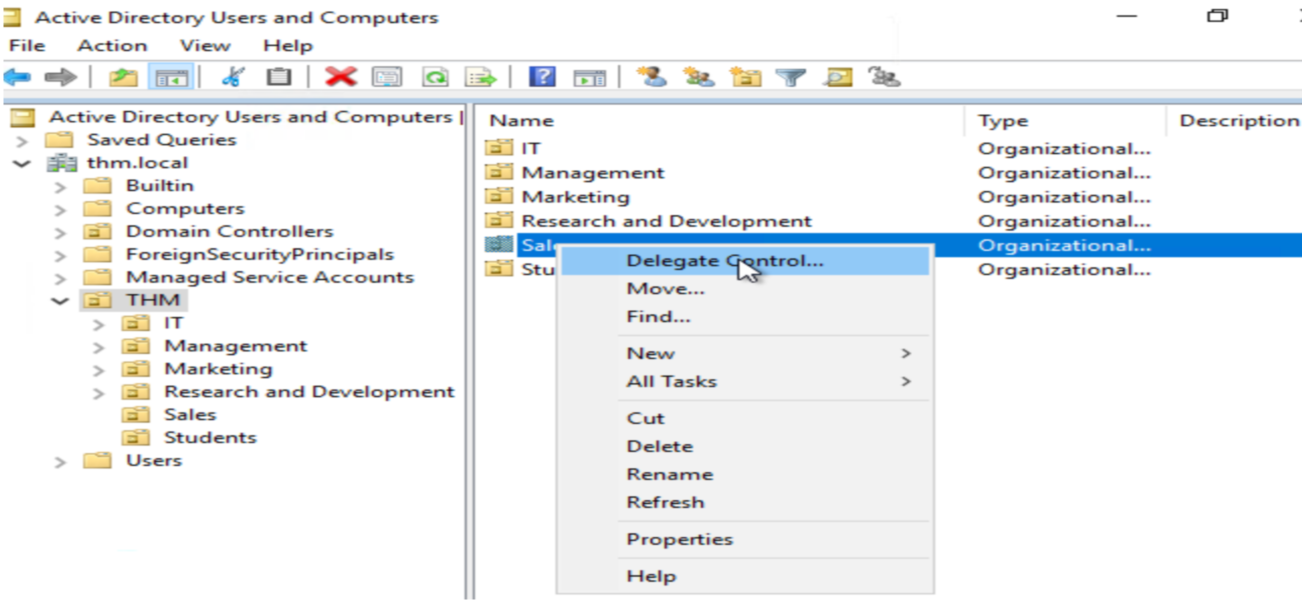
Security Groups vs Organization unit

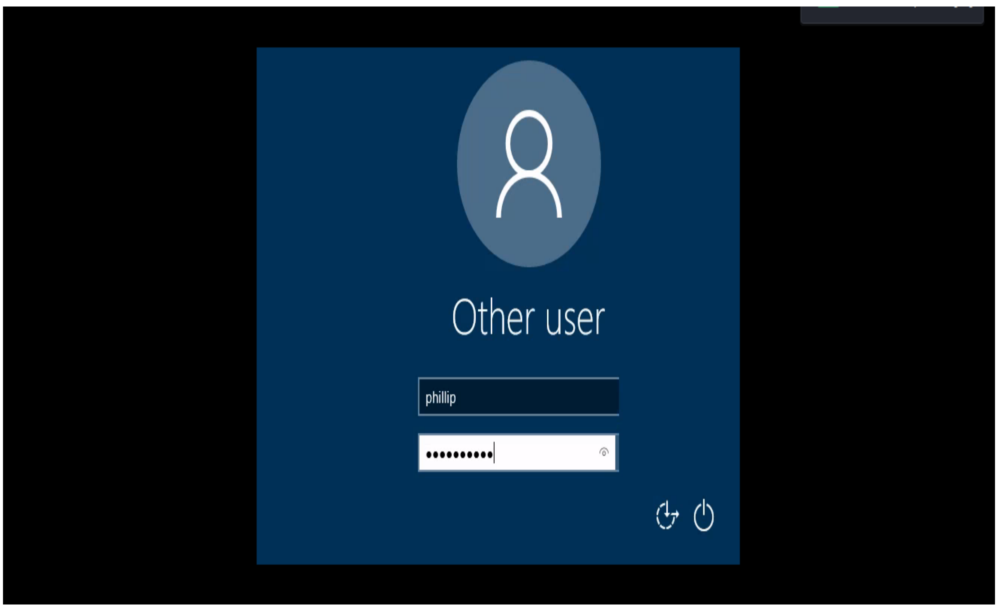
- **OUs** are for **applying policies** to users and computers, which include specific configurations that pertain to sets of users depending on their particular role in the enterprise. a user can only be a member of a single OU at a time, as it wouldn't make sense to try to apply two different sets of policies to a single user.

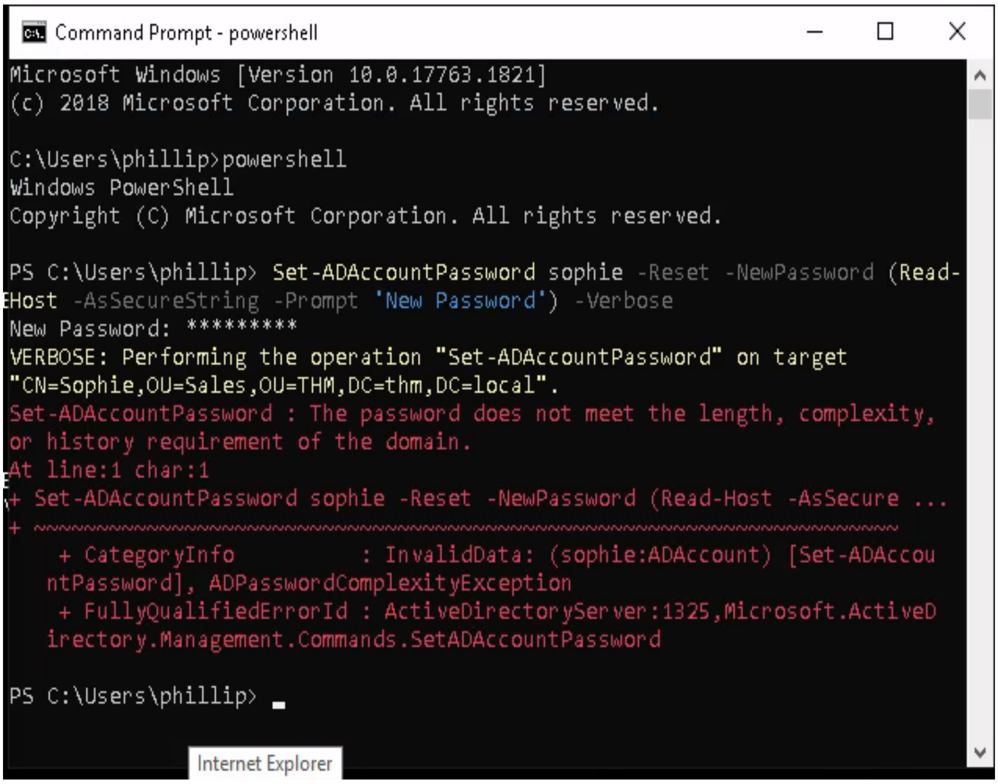
- **Security Groups**, are used to **grant permissions over resources**. For example, you will use groups if you want to allow some users to access a shared folder or network printer. A user can be a part of many groups, which is needed to grant access to multiple resources.

Managing Users in AD

Delegation- give specific users some control over some OUs.- allows you to grant users specific privileges to perform advanced tasks on OUs without needing a Domain Administrator to step in.





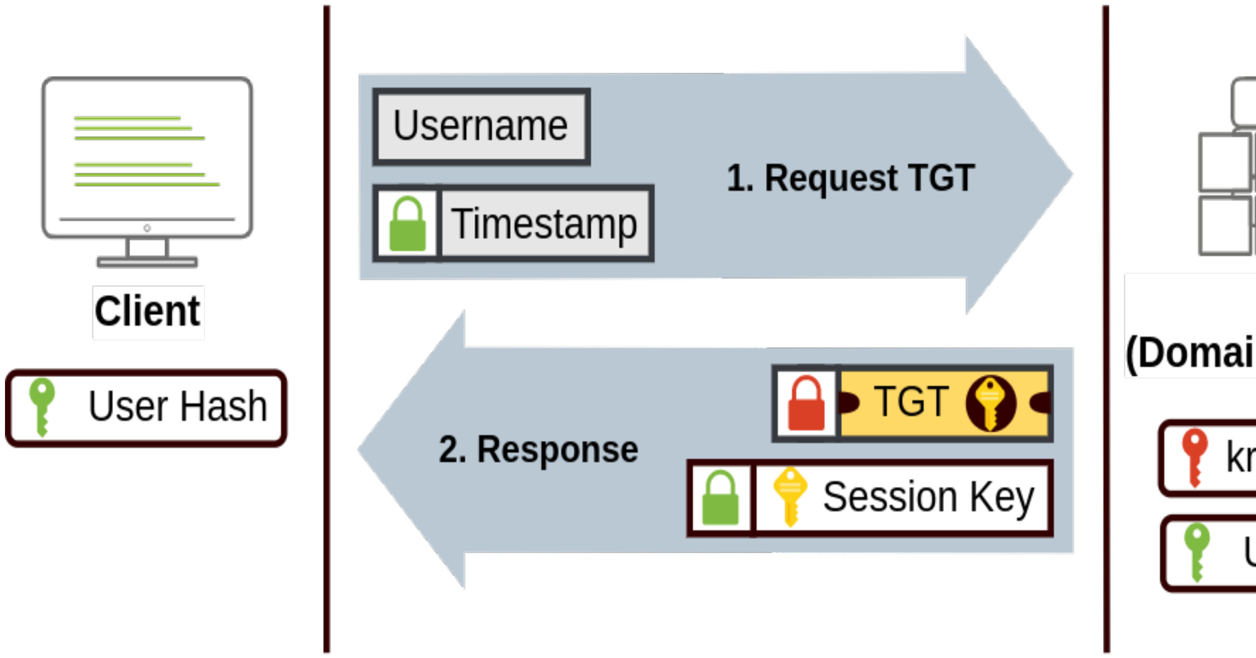


Authentication Methods

When using Windows domains, all credentials are stored in the Domain Controllers. Whenever a user tries to authenticate to a service using domain credentials, the service will need to ask the Domain Controller to verify if they are correct.

Two protocols can be used for network authentication in windows domains:

- **Kerberos:** Used by any recent version of Windows. This is the default protocol in any recent domain.
  How it works:



- **NetNTLM:** Legacy authentication protocol kept for compatibility purposes

  How it works